

Workshop 8 TEMPLATE

12pm

Callie Baker

2026-05-22

Table of contents

1. Set up	2
2. Clean data	3
3. Visualizing data	5
What is the relationship between SLA and plant mass?	5
What are the differences in total mass between water treatments?	6
What are the differences in total mass between species?	7
4. Fitting models	9
Model fitting	10
Model diagnostics	11
Model selection	12
Stop and think: what does this model mean?	13
4. Model predictions	14
5. Final figure	14
6. Writing	16
Figure caption	16
Basic components of test	16
Written communication	16
Methods	16
Results	16

In this workshop, we will answer the question: How do specific leaf area, water treatment, and species influence plant mass?

Specific leaf area is a continuous variable measured in cm^2/g .

Water treatment is a categorical variable (i.e. a *factor*) with 2 levels: drought stressed (DS) and well watered (WW).

Species is a categorical variable (again, a *factor*) with 6 levels.

Species name	Species code	Common name
<i>Encelia californica</i>	ENCCAL	Bush sunflower
<i>Eschscholzia californica</i>	ESCCAL	California poppy
<i>Penstemon centranthifolius</i>	PENCEN	Scarlet bugler
<i>Grindelia camporum</i>	GRINCAM	Gumweed
<i>Salvia leucophylla</i>	SALLEU	Purple sage
<i>Stipa pulchra</i>	STIPUL	Purple needlegrass
<i>Lotus scoparius</i>	LOTSCO	Deerweed

1. Set up

```
# read in your packages here
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(readxl) # reading excel files
library(scales) # modifying axis labels
library(ggeffects) # getting model predictions
library(ggplot2)
library(MuMIn) # model selection
library(dplyr)

# read in your data here

drought_exp <- read_xlsx(
  here("data", "Valliere_etal_EcoApps_Data.xlsx"),
  sheet = "First Harvest"
)
```

```
# storing colors to use for species
lotsco_col <- "#E69512"
pencen_col <- "#D6264F"
salleu_col <- "#6D397D"
```

```

enccal_col <- "#3A5565"
stipul_col <- "#3F564F"
esccal_col <- "#515481"
gricam_col <- "#6C91BD"

# storing colors to use for water treatments
ds_col <- "#A62F03"
ww_col <- "#045CB4"

# storing a ggplot theme (that will be used for all ggplots)
theme_set(theme_bw()+
  theme(text = element_text(size = 14)))

```

2. Clean data

Data source: Valliere, Justin; Zhang, Jacqueline; Sharifi, M.; Rundel, Philip (2019). Data from: Can we condition native plants to increase drought tolerance and improve restoration success? [Dataset]. Dryad. <https://doi.org/10.5061/dryad.v0861f7>

```

# created new object from drought_exp
drought_exp_clean <- drought_exp |>
  # makes column names snake case and replaces spaces with underscores
  clean_names() |>
  # creating new column with full scientific name
  mutate(species_name = case_match(
    species,
    "ENCCAL" ~ "Encelia californica", # bush sunflower
    "ESCCAL" ~ "Eschscholzia californica", # California poppy
    "PENCEN" ~ "Penstemon centranthifolius", # Scarlet bugler
    "GRICAM" ~ "Grindelia camporum", # Gumweed
    "SALLEU" ~ "Salvia leucophylla", # purple sage
    "STIPUL" ~ "Stipa pulchra", # purple needlegrass
    "LOTSCO" ~ "Lotus scoparius" # deerweed
  )) |>
  # creating new column with full treatment names
  mutate(water_treatment = recode_values(
    water,
    "WW" ~ "Well watered",
    "DS" ~ "Drought stressed"
  )) |>
  # reordering species name levels

```

```
mutate(species_name = as_factor(species_name),
       species_name = fct_relevel(species_name,
                                   "Lotus scoparius",
                                   "Penstemon centranthifolius",
                                   "Salvia leucophylla",
                                   "Encelia californica",
                                   "Stipa pulchra",
                                   "Eschscholzia californica",
                                   "Grindelia camporum")) |>

# reordering water_treatment levels
mutate(water_treatment = as_factor(water_treatment),
       water_treatment = fct_relevel(water_treatment,
                                      "Drought stressed",
                                      "Well watered")) |>

# selecting columns of interest
select(species_name, water_treatment, sla, total_g)
```

Insert code to double check the structure of the data frame here.

```
str(drought_exp_clean)
```

```
tibble [70 x 4] (S3: tbl_df/tbl/data.frame)
 $ species_name   : Factor w/ 7 levels "Lotus scoparius",...: 4 4 4 4 4 4 4 4 4 4 ...
 $ water_treatment: Factor w/ 2 levels "Drought stressed",...: 2 2 2 2 2 1 1 1 1 1 ...
 $ sla            : num [1:70] 170 215 209 216 222 ...
 $ total_g        : num [1:70] 0.455 0.329 0.45 0.359 0.352 ...
```

Insert code to display 10 random rows from the data frame here.

```
slice_sample(
  drought_exp_clean, # cleaned data frame to display
  n = 10 # showed ten rows
)
```

```
# A tibble: 10 x 4
```

species_name	water_treatment	sla	total_g
<fct>	<fct>	<dbl>	<dbl>
1 Salvia leucophylla	Well watered	96.5	0.4
2 Grindelia camporum	Drought stressed	171.	0.250
3 Encelia californica	Well watered	209.	0.450

4	Salvia leucophylla	Well watered	96.2	0.232
5	Penstemon centranthifolius	Well watered	129.	0.220
6	Stipa pulchra	Drought stressed	176.	0.181
7	Salvia leucophylla	Well watered	113.	0.195
8	Stipa pulchra	Well watered	161.	0.376
9	Lotus scoparius	Drought stressed	116.	0.0802
10	Penstemon centranthifolius	Drought stressed	175.	0.188

3. Visualizing data

What is the relationship between SLA and plant mass?

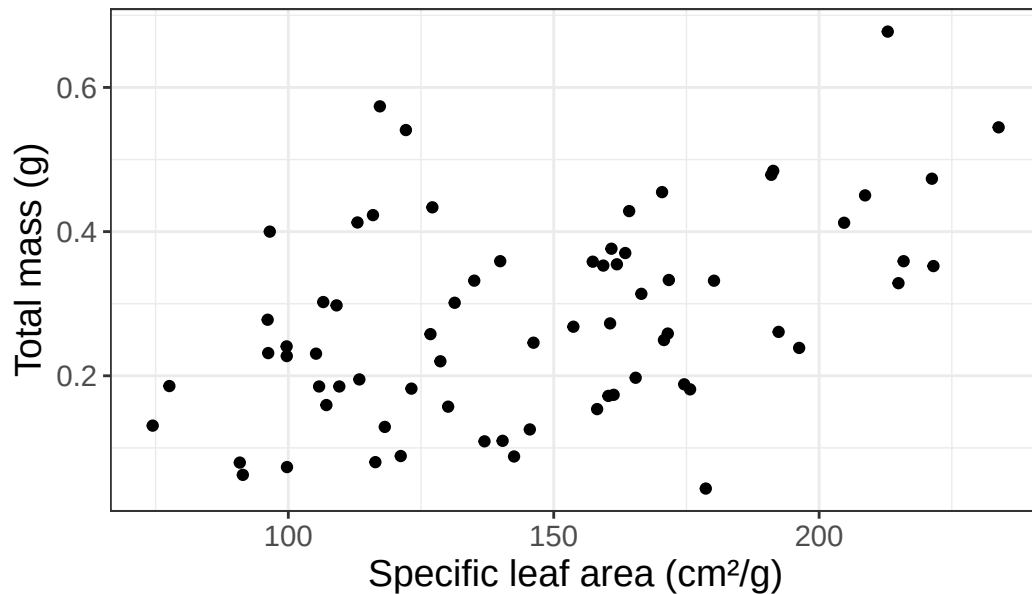
Make a scatterplot to visualize SLA (the predictor) on the x-axis, and total mass (the response) on the y-axis.

Label the axes with units where appropriate.

Add a title describing the relationship (e.g. as SLA increases, total mass _____).

```
# annotate
ggplot(data = drought_exp_clean,
       # x-axis: SLA
       # y-axis: total plant mass
       mapping = aes(x = sla,
                     y = total_g)) +
  # annotate
  geom_point() +
  # annotate
  labs(title = "As SLA increases, total mass increases",
       x = "Specific leaf area (cm2/g)",
       y = "Total mass (g)") +
  # changed theme
  theme(legend.position = "none")
```

As SLA increases, total mass increases



What are the differences in total mass between water treatments?

Make a jitterplot to visualize the differences in plant mass between water treatments.

Label the axes with units where appropriate.

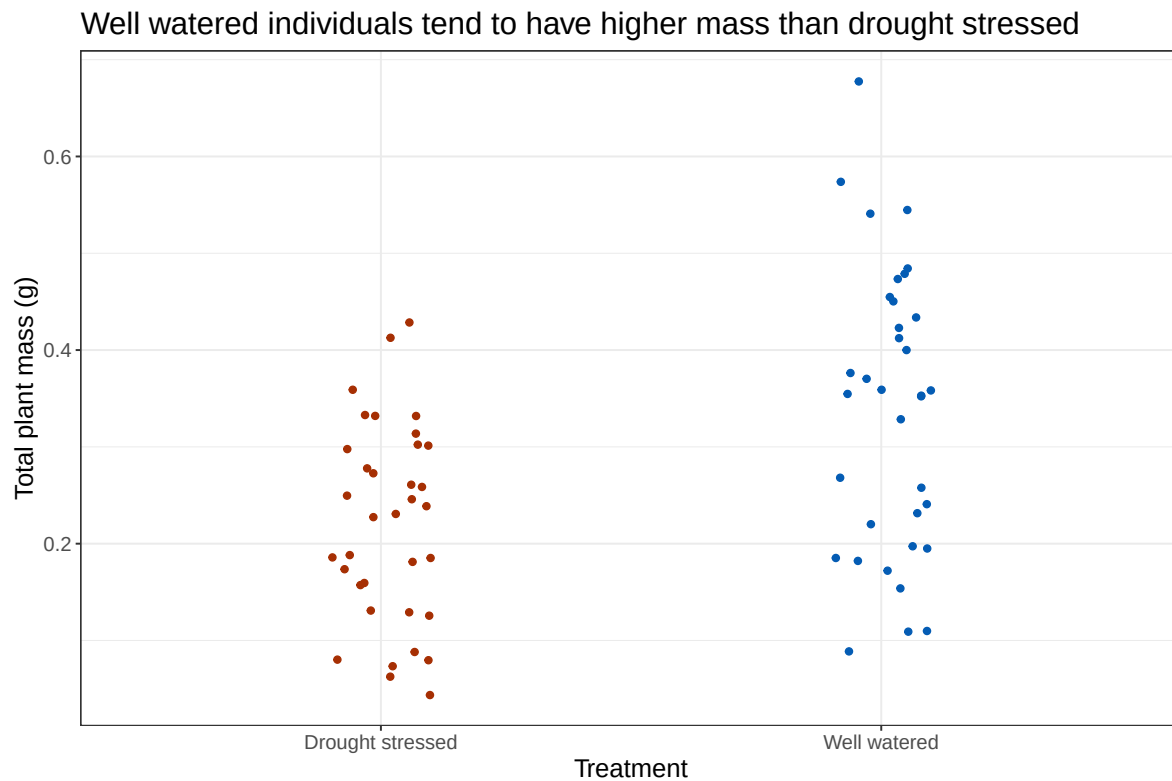
Add a title describing the differences in (mean or median) plant mass between water treatments.

```
# base layer: ggplot
ggplot(data = drought_exp_clean,
  # x-axis: water treatment
  # y-axis: total mass
  # colored geometries by water treatment
  mapping = aes(x = water_treatment,
    y = total_g,
    color = water_treatment)) +
  # representing individuals
  geom_jitter(height = 0,
    width = 0.1) +
  # custom colors
  scale_color_manual(values = c(
    "Drought stressed" = ds_col,
```

```

  "Well watered" = ww_col)) +
# added title and relabeled axes
labs(title = "Well watered individuals tend to have higher mass than drought stressed",
     x = "Treatment",
     y = "Total plant mass (g)") +
# removed legend
theme(legend.position = "none")

```



What are the differences in total mass between species?

Make a jitterplot to visualize the differences in plant mass between species.

Label the axes with units where appropriate.

Add a title describing the differences in (mean or median) plant mass between species.

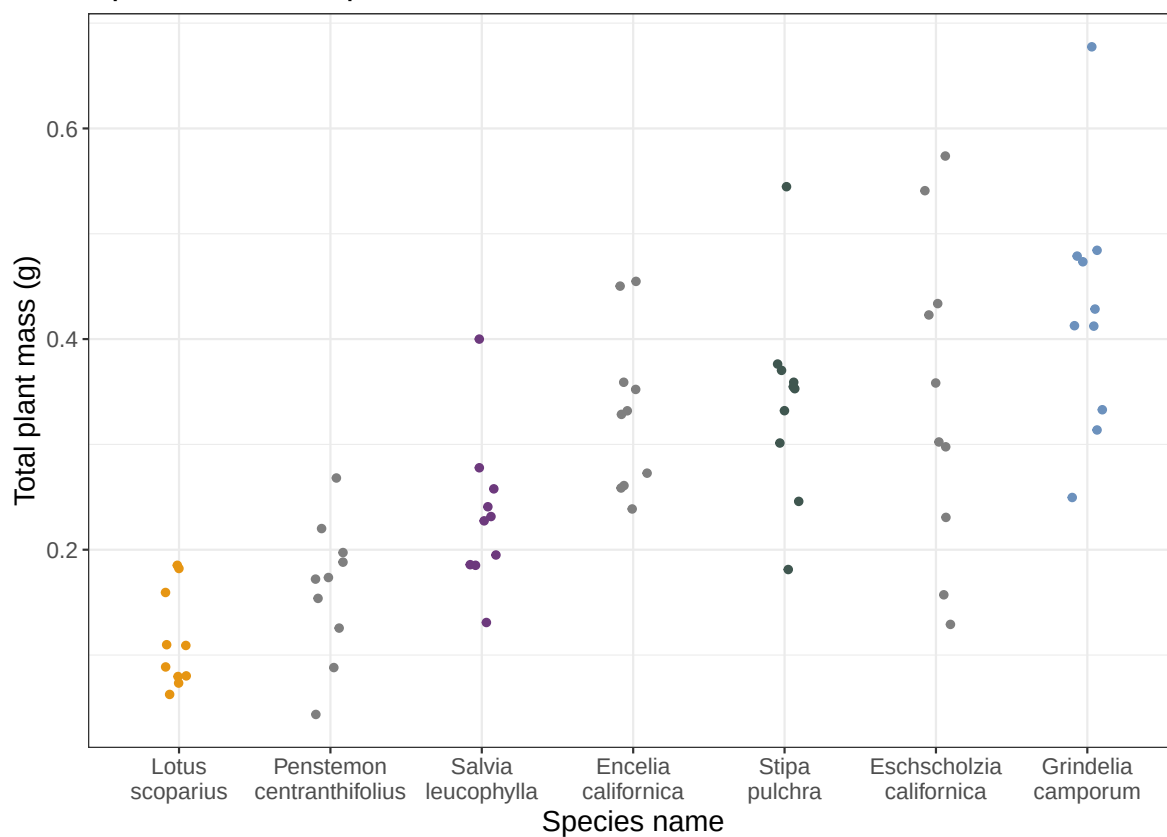
Bonus: order the x-axis by decreasing mean plant mass.

```

# base layer: ggplot
ggplot(data = drought_exp_clean,
      # x-axis: species
      # y-axis: plant mass
      # colored by species name
      mapping = aes(x = species_name,
                    y = total_g,
                    color = species_name)) +
# representing individuals in each species
geom_jitter(height = 0,
            width = 0.1) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(10)) +
# added title and relabeled axes
labs(title = "Species differ in plant mass",
      x = "Species name",
      y = "Total plant mass (g)") +
# using custom colors
scale_color_manual(
  values = c(
    "Lotus scoparius" = lotsco_col,
    "Pentsemon centranthifolius" = pencen_col,
    "Salvia leucophylla" = sallev_col,
    "Encelia californica" = enccal_col,
    "Stipa pulchra" = stipul_col,
    "Eschscholzia californica" = esccal_col,
    "Grindelia camporum" = gricam_col)) +
# removed legend
theme(legend.position = "none")

```


Species differ in plant mass



4. Fitting models

8 models total:

Model number	SLA	Water treatment	Species	Predictor list
0				no predictors (null model)
1	X	X	X	all predictors (full model)
2	X	X		SLA and water treatment
3	X		X	SLA and species
4		X	X	water treatment and species
5	X			SLA
6		X		water treatment
7			X	species

Model fitting

```
# model 0: null model
model0 <- lm(
  total_g ~ 1,
  data = drought_exp_clean)

# model 1: all predictors
model1 <- lm(
  total_g ~ sla + water_treatment + species_name,
  data = drought_exp_clean)

# model 2: SLA and water treatment
model2 <- lm(
  total_g ~ sla + water_treatment,
  data = drought_exp_clean)

# model 3: SLA and species
model3 <- lm(
  total_g ~ sla + species_name,
  data = drought_exp_clean)

# model 4: water treatment and species
model4 <- lm(
  total_g ~ water_treatment + species_name,
  data = drought_exp_clean)

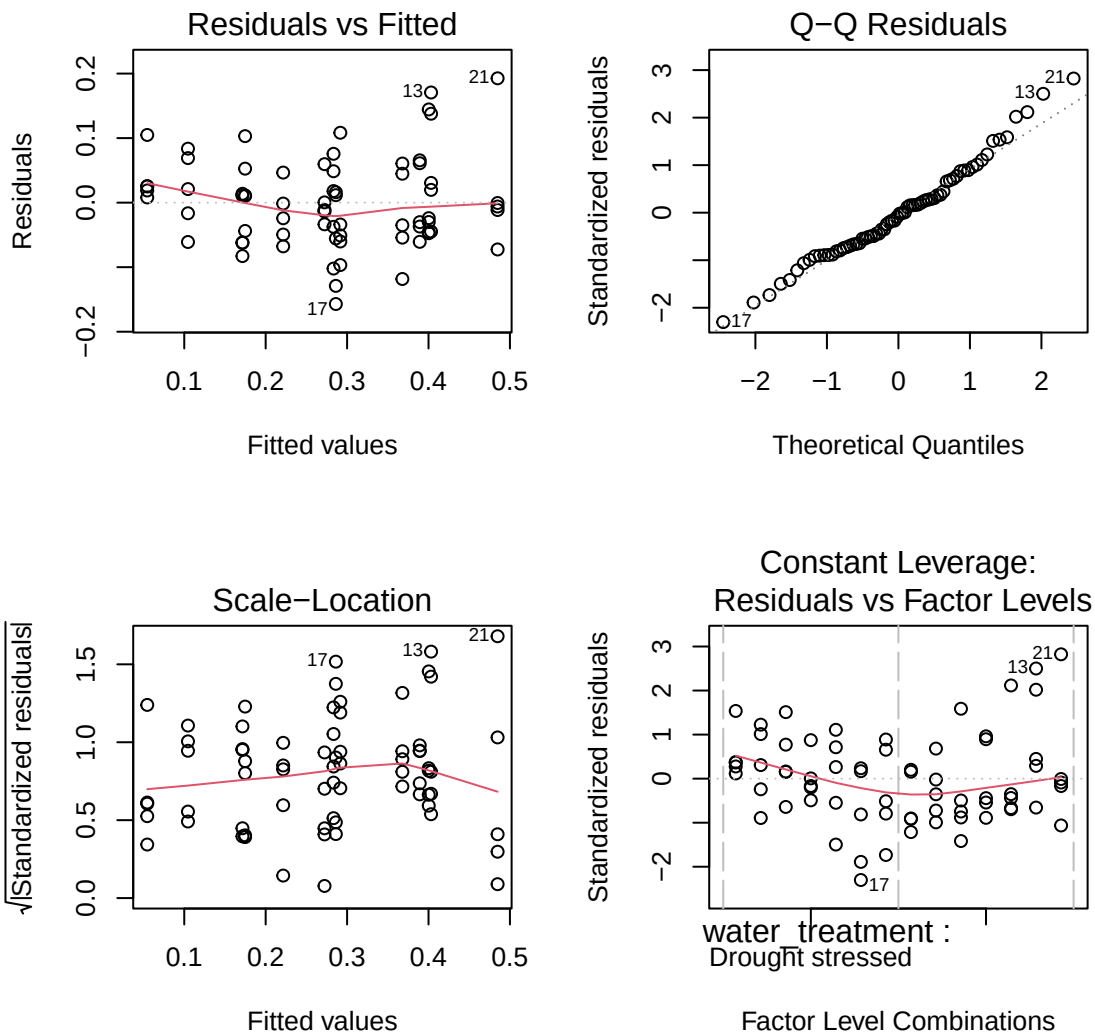
# model 5: SLA
model5 <- lm(
  total_g ~ sla,
  data = drought_exp_clean)

# model 6: water treatment
model6 <- lm(
  total_g ~ water_treatment,
  data = drought_exp_clean)

# model 7: species
model7 <- lm(
  total_g ~ species_name,
  data = drought_exp_clean)
```

Model diagnostics

```
#displaying lots in 2x2 grid
par(mfrow = c(2,2))
# plotting diagnostics
plot(model14)
```



- residuals are normally distributed (QQ plot)
- residuals are homoscedastic (Residuals vs Fitted/Scale-Location)

- no outliers that could be influencing model coefficient estimates (Leverage)

Model selection

Insert code to use the `AICc()` function to calculate the AIC for each model.

Arrange by decreasing AIC.

```
AICc(
  model0,
  model1,
  model2,
  model3,
  model4,
  model5,
  model6,
  model7
) |>
  arrange(AICc)
```

	df	AICc
model4	9	-156.19595
model11	10	-153.75361
model3	9	-124.07569
model7	8	-120.30191
model2	4	-95.82521
model5	3	-88.85180
model6	3	-86.76661
model0	2	-74.98036

Insert code to display the summary for the best model.

```
summary(model4)
```

Call:

```
lm(formula = total_g ~ water_treatment + species_name, data = drought_exp_clean)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.157087	-0.046953	-0.003733	0.041244	0.192657

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.05455	0.02451	2.225	0.02973 *
water_treatmentWell watered	0.11695	0.01733	6.746	5.90e-
09 ***				
species_namePenstemon centranthifolius	0.05003	0.03243	1.543	0.12799
species_nameSalvia leucophylla	0.12020	0.03243	3.706	0.00045 ***
species_nameEncelia californica	0.21774	0.03243	6.714	6.70e-
09 ***				
species_nameStipa pulchra	0.22881	0.03243	7.055	1.72e-
09 ***				
species_nameEschscholzia californica	0.23164	0.03243	7.143	1.22e-
09 ***				
species_nameGrindelia camporum	0.31335	0.03243	9.662	5.53e-
14 ***				

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07252 on 62 degrees of freedom

Multiple R-squared: 0.7535, Adjusted R-squared: 0.7257

F-statistic: 27.08 on 7 and 62 DF, p-value: < 2.2e-16

Stop and think: what does this model mean?

What is the best model?

The best model includes water treatment and species as predictors

How much variation in the response (total mass, in grams) does this model explain?

72.6% of the variation in plant mass is explained by this model.

How do we interpret the effects of the predictors on the response variable (again, total mass in grams)?

- species influences plant mass
- treatment influences plant mass
- (Intercept): The intercept is the model prediction for plant mass for Lotus scoparius in the drought stressed treatment.
- water_treatmentWell watered: The model prediction for the difference in Lotus scoparius mass between well watered and drought stressed treatments.

- `species_namePenstemon centranthifolius`: The model prediction for the difference between *Penstemon* mass in the well water treatment and *Lotus scoparius* mass in the well watered treatment.
- `species_nameEschscholzia californica`: difference between *Eschscholzia californica* mass in the well watered treatment and *Lotus scoparius* mass in the well watered treatment.

4. Model predictions

Insert code to generate model predictions for the best model.

Rename the `x` column and `group` column to match the original data frame (`drought_exp_clean`).

Look at the object before plotting.

```
preds <- ggpredict(
  model4, # model object
  terms = c("water_treatment", "species_name")) |> # predictors
# renamed column to fit original data frame
rename(water_treatment = x,
       species_name = group)
```

5. Final figure

Create a clean, publication ready figure depicting predictions for the best model and the underlying data.

```
# base layer: ggplot
ggplot(data = drought_exp_clean,
       mapping = aes(x = water_treatment,
                     y = total_g,
                     color = water_treatment)) +

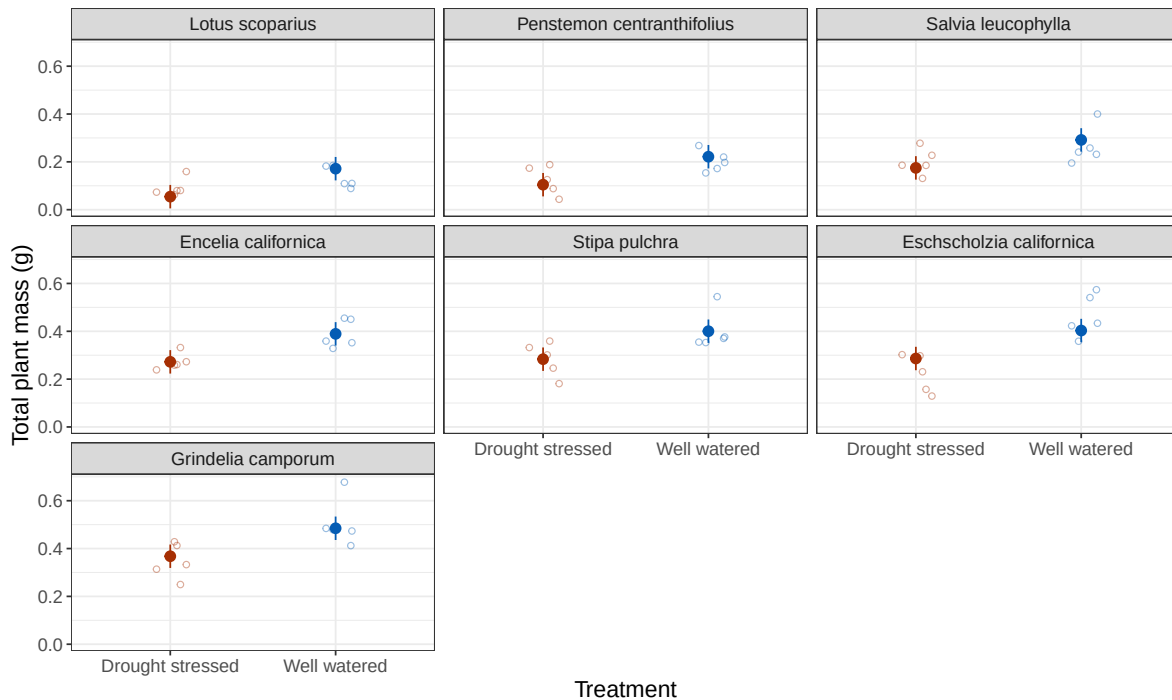
# added jitter
geom_jitter(height = 0,
            width = 0.1,
            alpha = 0.4,
            shape = 21) +

# model predictions with 95% CI
geom_pointrange(data = preds,
               mapping = aes(x = water_treatment,
```

```

    y = predicted,
    ymin = conf.low,
    ymax = conf.high)) +
# creates different panels for each species
facet_wrap(~ species_name) +
# adds custom colors
scale_color_manual(values = c(
  "Drought stressed" = ds_col,
  "Well watered" = ww_col
)) +
# removes legend
theme(legend.position = "none") +
# relabeled axes
labs (x = "Treatment",
      y = "Total plant mass (g)")

```



6. Writing

Figure caption

Basic components of test

predictors

response

test

distribution

parameters (degrees of freedom)

R^2

p-value

significance level

reference level(s)

Written communication

Methods

We used linear models to determine the predictive relationship between specific leaf area (cm^2/g), treatment, and species on plant mass (g). We evaluated model assumptions of normally distributed errors using QQ plots of model residuals. We evaluated assumptions of constant variance of errors using Residuals vs Fitted and Scale-Location plots. We used Akaike's Information Criterion (AIC) to select the simplest model that explained the most variation in plant mass.

Results

We found that the simplest model describing the greatest variation in plant mass included treatment and species as predictors (linear model: $F(2, 67) = 14.57$, $p < 0.001$, $\alpha = 0.05$).